

Applications of AI for Engineering, Science and Writing



Be curious.

Make magic.

"If you go back a few hundred years, what we take for granted today would seem like magic- being able to talk to people over long distances, to transmit images, flying, accessing vast amounts of data like an oracle. These are all things that would have been considered magic a few hundred years ago." -Elon Musk

4 Historical “De-Centerings”

Copernican Revolution: Galileo Galilei made us realize we are not the center of the universe.

Darwinian Revolution: Charles Darwin made us realize we aren't the pinnacle of evolution.

Freudian Revolution: Sigmund Freud made us realize we don't know ourselves.

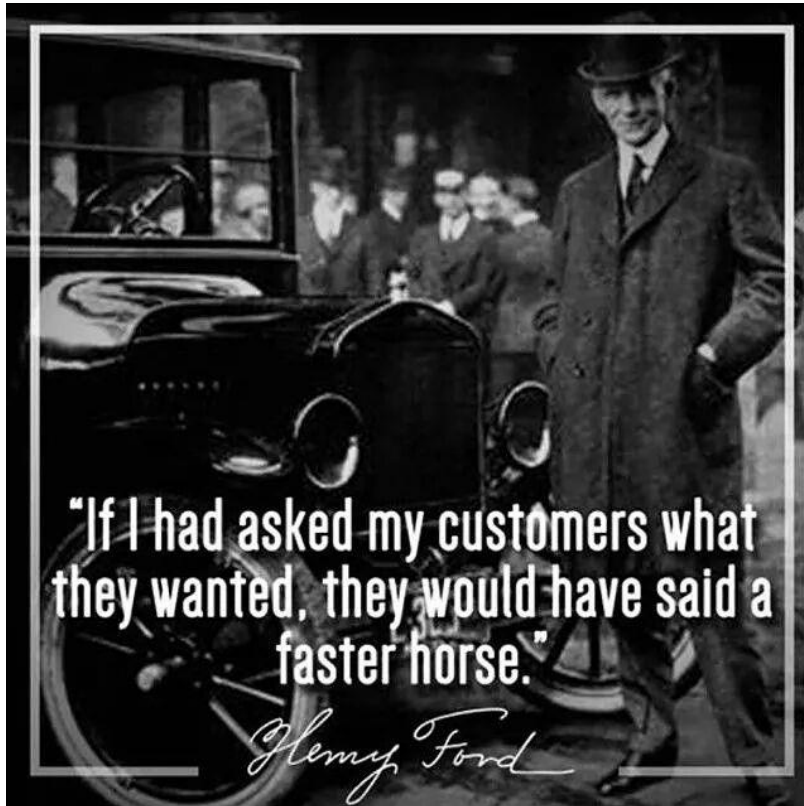
The Fourth Revolution: Artificial Intelligence will make us realize we are not that smart.

This ongoing revolution of AI and the "infosphere" is displacing us from the center of the world of information and intelligence. It challenges the long-held belief in the uniqueness and superiority of human reason.

Luciano Floridi, [The Fourth Revolution: How the Infosphere is Reshaping Human Reality](#), 2014

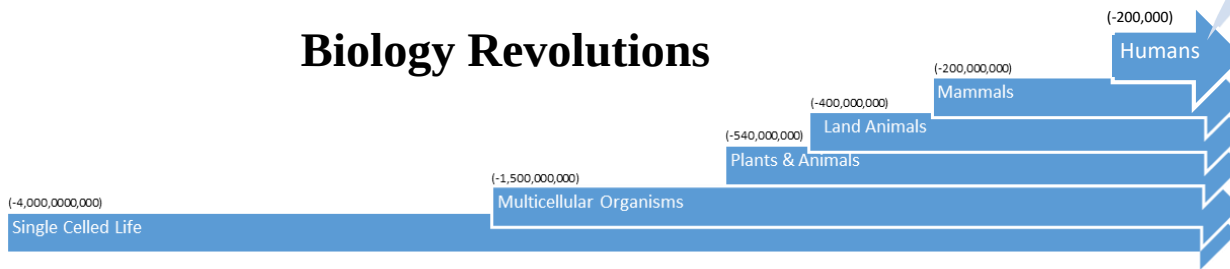
Disruptive Technology

An innovation that creates a *new* market by providing a different set of values, which ultimately (and unexpectedly) overtakes an existing market (e.g., the lower-priced, affordable [Ford Model T](#), which displaced horse-drawn carriages)

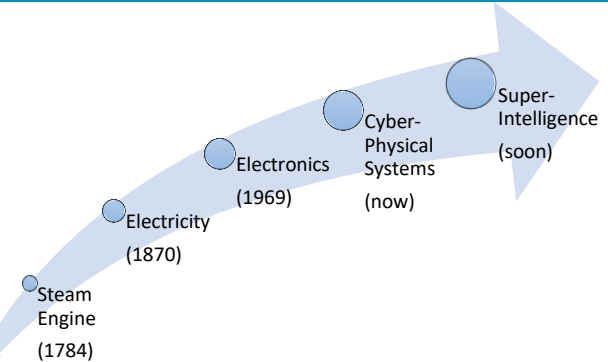


Biomedical Engineering

Biology Revolutions



Technology Revolutions



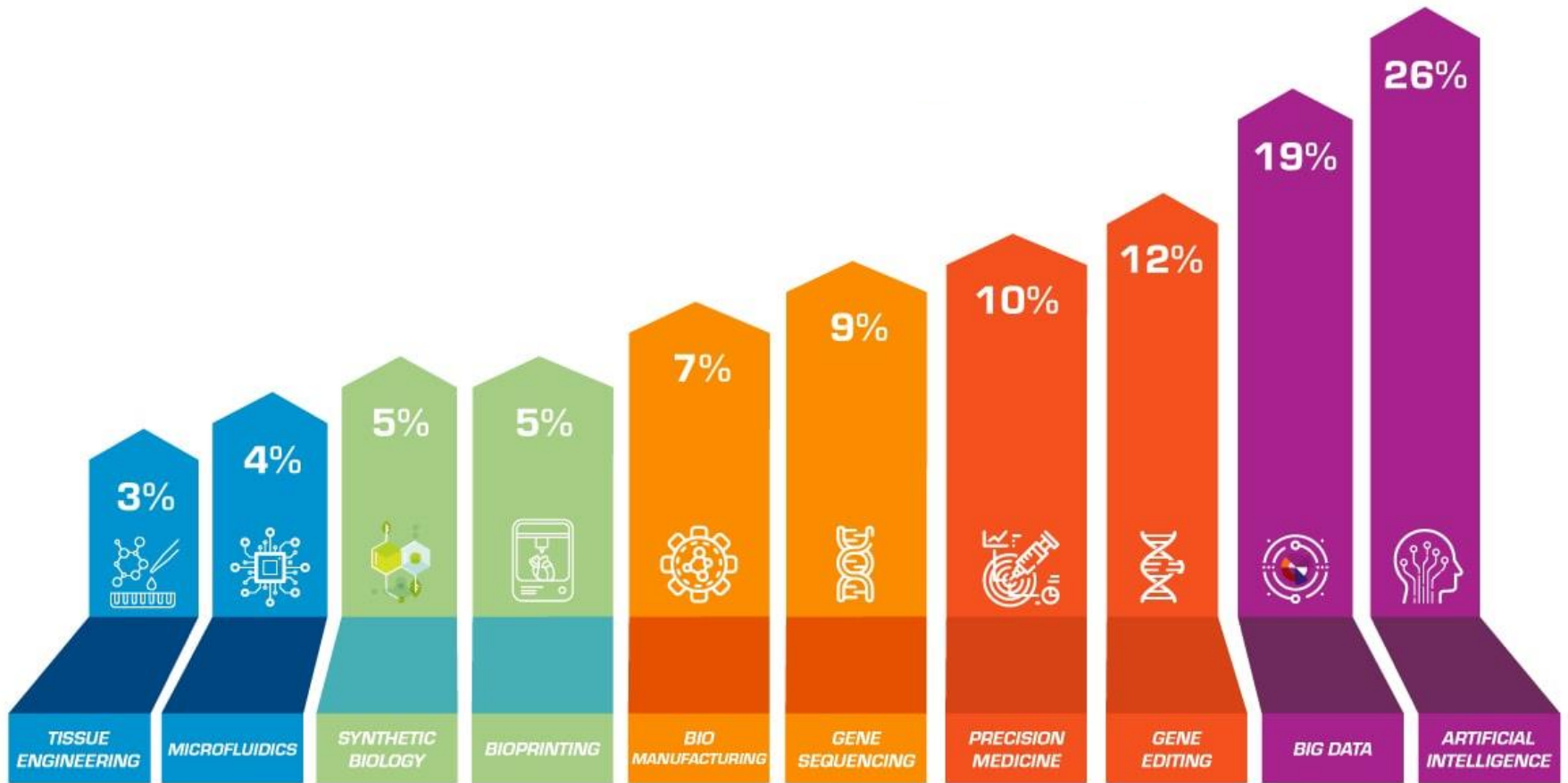
INDUSTRY 4.0

Mobile
Cloud
Big data
Sensors
IOT



AI
Digital
Simulation
Social
Crypto

Biotechnology Trends



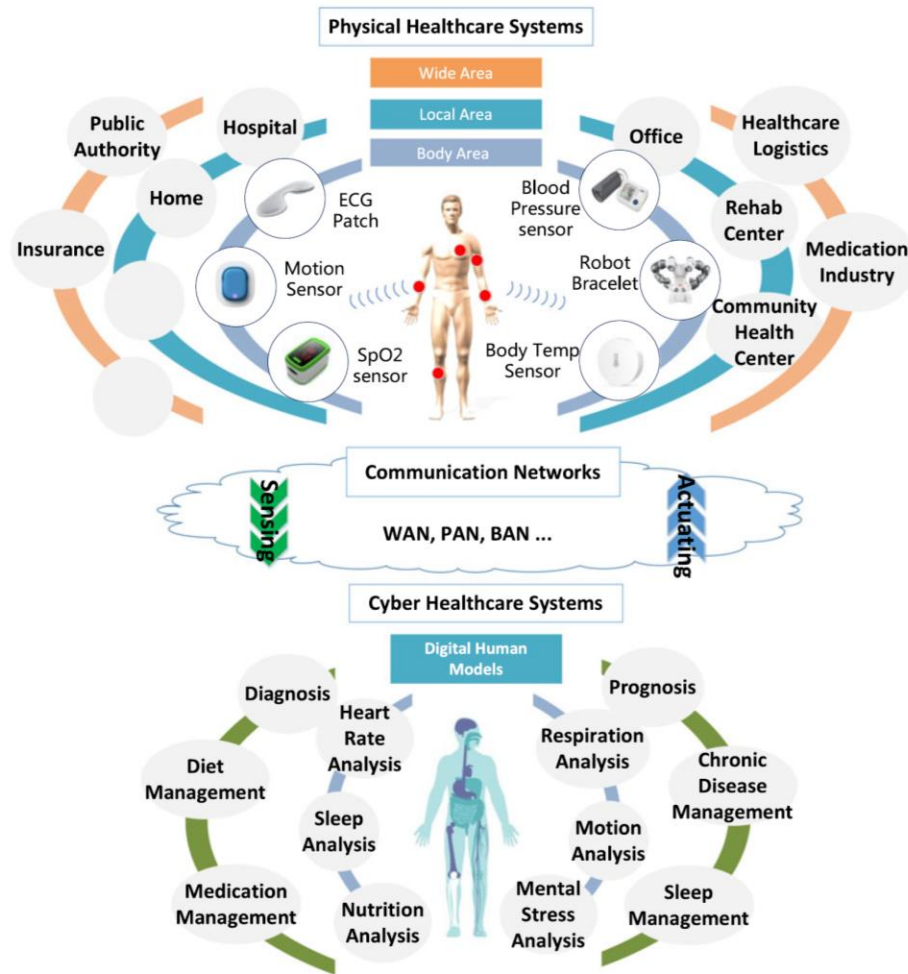
Source: StartUs Insights January 2022

“Personalized” Medicine



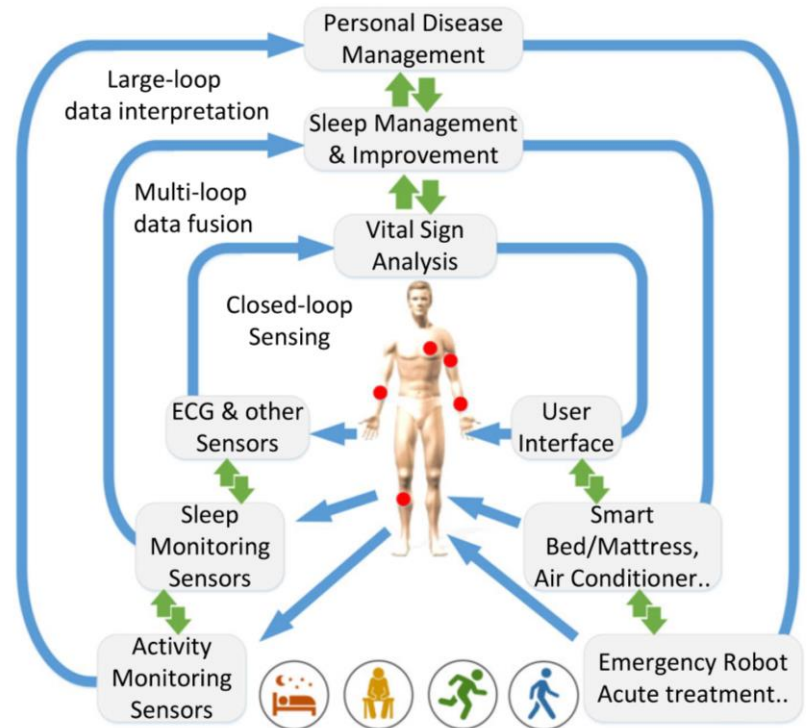
Schüssler-Fiorenza Rose, S.M., Contrepois, K., Moneghetti, K.J. *et al.* A longitudinal big data approach for precision health. *Nat Med* **25**, 792–804 (2019).

Healthcare 4.0



Paradigm shift in Caregiving:

- From open loop to closed loop
- From single loop to multiple loops
- From small loop to large loop



Pang, Zhibo, et al. "Introduction to the special section: convergence of automation technology, biomedical engineering, and health informatics toward the healthcare 4.0." *IEEE Reviews in Biomedical Engineering* 11 (2018): 249-259.

Opportunities

ENGINEERING UNLEASHED

POWERED BY KEEN



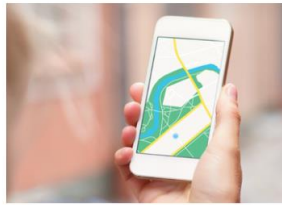
Curiosity

Understand the broader world. Look forward. Explore multiple perspectives.



Connections

Think outside the box. Place old ideas in new contexts. Gain insights.

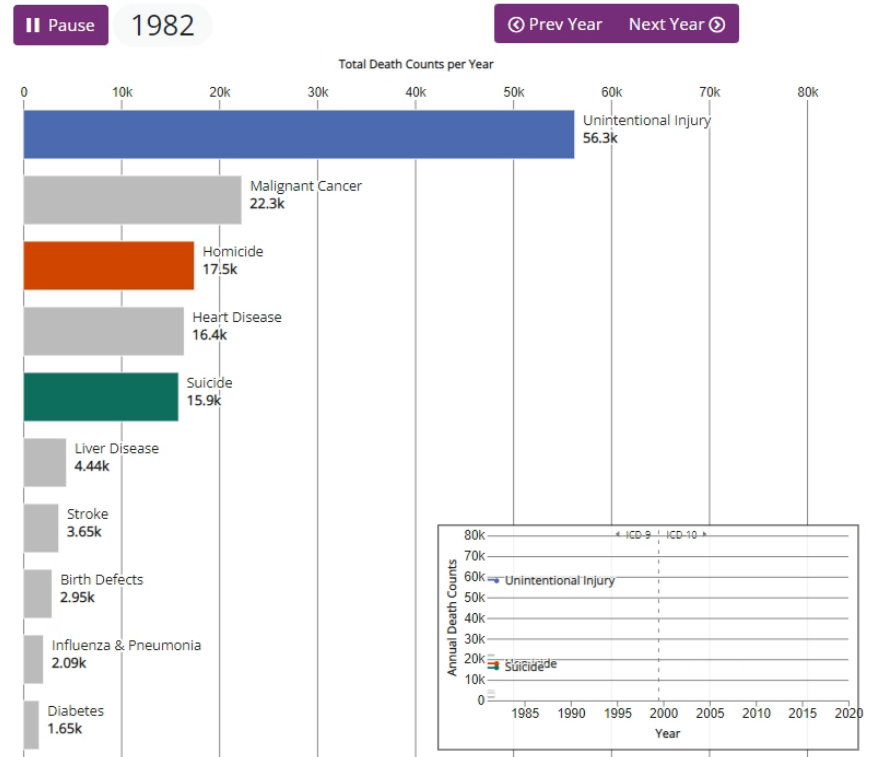


Creating Value

Seek opportunities. Understand stakeholders. Have a positive impact.



MINDSET + SKILLSET
A TANDEM DEVELOPMENT



www.cdc.gov/injury/wisqars/

AI-Enabled Toolkits

I am creating a lecture for engineering students in a design class and for my slide on open/free toolkits that are available to startups/entrepreneurs I would like to update it with the new revolutionary tools that AI has enabled in the past few years. Can you provide some general themes and specific tools that they will find useful? Convert the summary recommendations into a bulleted list graphic that I can add to my slide in the lecture.

Revolutionary AI-Enabled Toolkits for Startups.

Generative Design
Surrendering to the brutal fact that in our culture, we are a city of the future, it is our duty to create.

Automation
Confrontation is the only solution to the problem of the future.

Content Creation
The only way to survive in the future is to be the first to see the future.

Data Insights
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Open-Source AI Frameworks
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Sustainable Development
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Featured Tools

Loveable

Generative Design

Automation

Content Creation

Data Insights

Open-Source AI Frameworks

Canva AI

Generative Design

Automation

Content Creation

Data Insights

Open-Source AI Frameworks

ChatGPT

Generative Design

Automation

Content Creation

Data Insights

Open-Source AI Frameworks

TensorFlow

Generative Design

Automation

Content Creation

Data Insights

Open-Source AI Frameworks

Why It Matters

The only way to survive in the future is to be the first to see the future.

<https://x.com/i/grok/share/l8kzmKY9xQ0SDXrN9lFYGg7PO>

Content & Marketing

- ChatGPT: Brainstorm & Write
- Copy.ai: Social Posts & Ads

Development & Apps

- Copilot: Code Assistant
- Zoho: No-Code Apps

Change: Data Analytics

er Support & CRM

pot: Lead Management

24: Workflow CRM

nsights

L: AI Search Platform

ush: SEO Analysis

on & Ideation

ras: Idea Feedback

.ai: Slide Decks

AI Toolkits for Startups



Ideation & Research
(Stratup.ai, Perplexity)



Design & Prototyping
(Uizard, Leo Liding)



Content & Marketing
(ChatGPT, Canva)



Software Development
(GitHub Copilot, TensorFlow)



Business & Automation
(Startuptools.ai, Zapier)



<https://g.co/gemini/share/35687ce16f18>

Engineering Pedagogy



BME 3303 “Introduction to Biomechanics”

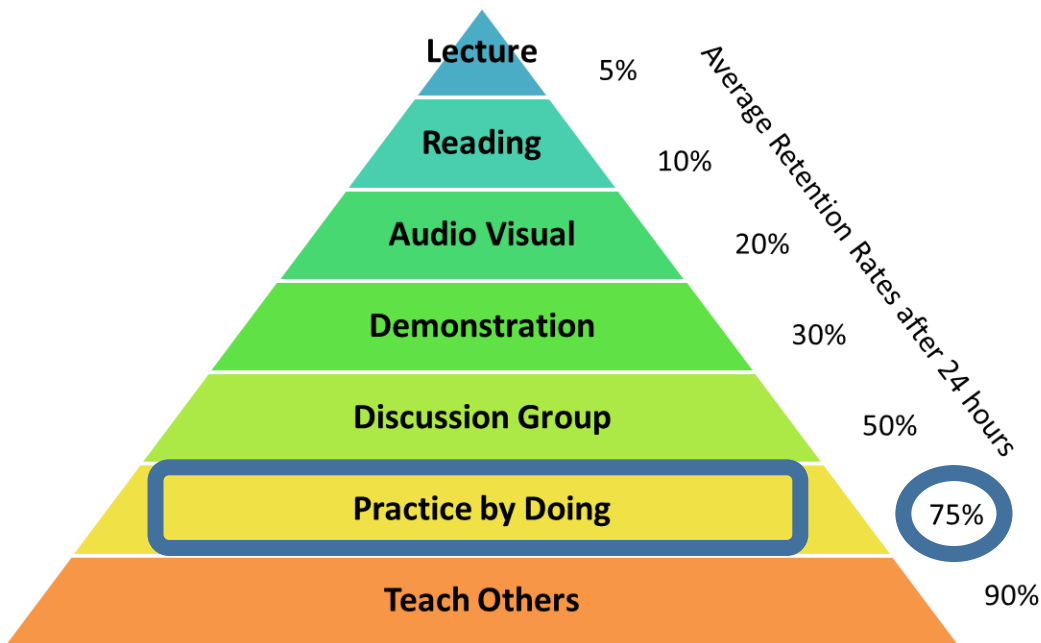
The LTU Softball coach has contacted you because of your expertise in biomechanics...





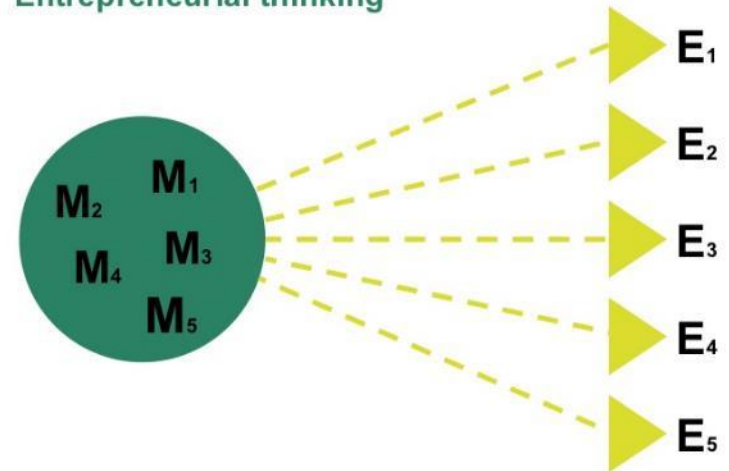
Learning by Doing and EML

The Learning Pyramid



Source: National Training Laboratories, Bethel Maine

Effectual reasoning Entrepreneurial thinking

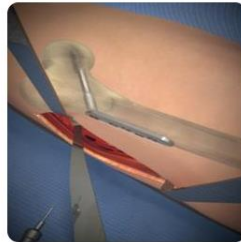


[Saras D. Sarasvathy](#) University of Virginia

Orthopedics Simulation Based Learning

- Simulation App
- Video Analysis
- Build-Break-Fix Models
- VR Demonstrations
- Prototyping
- Tissue Testing

Medtronic
Touch Surgery™ Enterprise



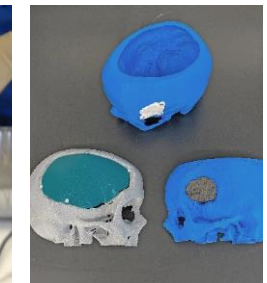
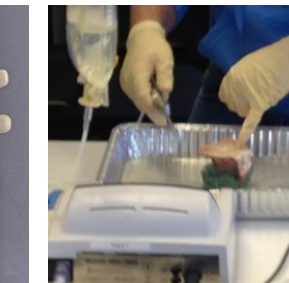
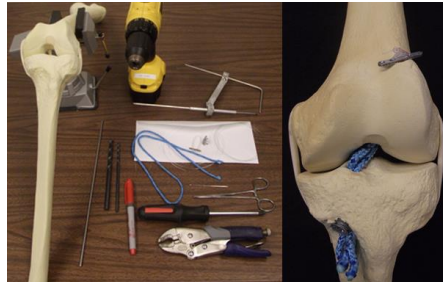
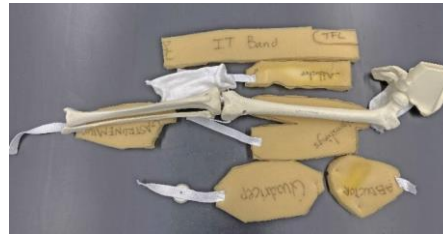
AO Stabilization using the Dynamic Hip Screw

Specialty
Orthopedics and Trauma

Release Date
March 15, 2018

Institution
AO Foundation

[Launch in App](#)



BME 5313

Engineering Applications in Orthopedics

- Weekly Journal Club
- Peer-reviewed Research Paper
- Opportunity Creation Project
- Medical Devices Regulation
- Orthopedics Industry
- Clinical/Hospital Tours

BME 5305 Orthopedics Take Home Final

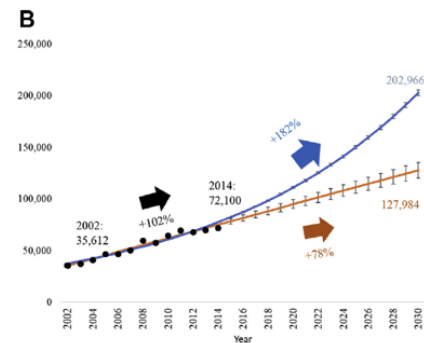
Due: Thursday 5/8/2025 3:45 pm

For the following questions, use generative AI (Perplexity, NotebookLM, or others) to suggest topics of interest and comparison reference articles. Submit the link to your AI discussion as well as **your own essay responses** (paragraph formatting) that includes adequate explanation, background information, referenced sources, and reflection on the impact of Biomedical Engineering for each question. The total length of your responses to the questions should be 2-3 pages each.

Generative AI Discussion Links (25 points) - be sure to follow-up on specific details, reference information or ask for review of other information sources (uploaded documents).

1. (25 pts) The chart below (from 2020) shows projected ranges for the number of revision total knee arthroplasty procedures expected through 2030 from <https://www.sciencedirect.com/science/article/pii/S0883540320301911>

Modern technology, such as ultra-high molecular weight polyethylene, newer tibial polyethylene geometries, component design to improve prosthetic stability, additives such as vitamin E and advancements in bearing surfaces may increase success rates, however even modern implants and well-fixed components have a finite life span that is more likely to be exceeded in the younger active patient.



Based on the topics discussed this semester and your knowledge from Biomedical Engineering, **describe one design constraint** for total knee arthroplasty and **how this impacts the number of revision surgeries**.

Grip Strength Quantification

Use the Hand Dynamometer to measure grip strength on both hands (3 trials for each). [Handexer Dynamometer](#)



Data Analysis



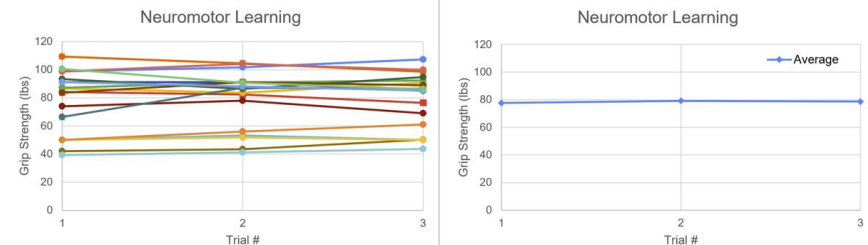
- Add grip strength measurements to the Google Form [Strength Data Collection](#)
- Download the Google Sheets with all measurement data [Strength Results](#)
- How can you analyze these data to create interesting and useful insights?
- Use prior knowledge from statistics or find a data analysis technique that you think will be helpful to understand these results. Create a professional graphical representation (graph, chart, table, diagram, etc...) that explains your analysis. Be sure to include a brief description with the graphic
- Compare your results with scientific references.

Discussion Board Post



Here is an example (based on limited preliminary data).

My hypothesis was that people would improve their grip strength by learning to control their hand after each trial. So I looked at the trends between Trial 1 to Trial 2 to Trial 3. The left graph shows all the different subjects and then I averaged each trial for the right graph. There did not seem to be an improvement from trial 2 to trial 3 and the difference between trial 1 and trial 2 was less than I expected.

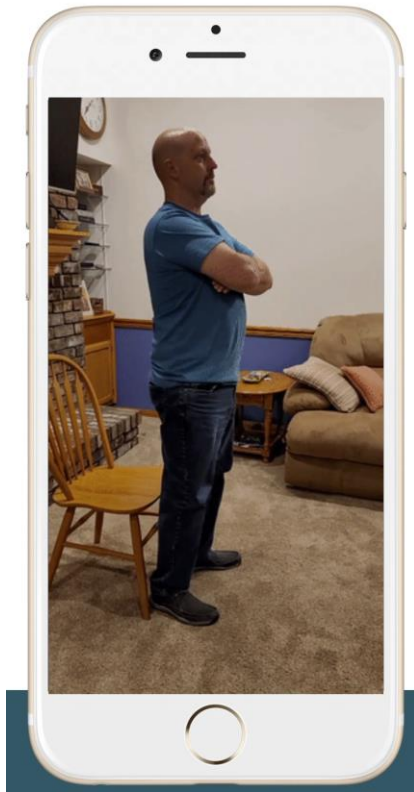


I asked Perplexity.ai "How does neuromuscular control and learning affect grip strength," and there were some good explanations in the references that support my hypothesis, such as; Neural Factors, Motor Unit Recruitment and Firing Rate, Intramuscular Coordination, Antagonist Inhibition, Skill Acquisition, Movement Pattern Optimization, and Proprioception.

<https://www.perplexity.ai/search/how-does-neuromuscular-control-R8vgE0.HSfeid.KF0TaWAag>

Computer Vision Biomechanics

- Use smartphone camera to record 5 cycle sit-to-stand movements.
- Upload video to sit2stand.ai for motion capture analysis.

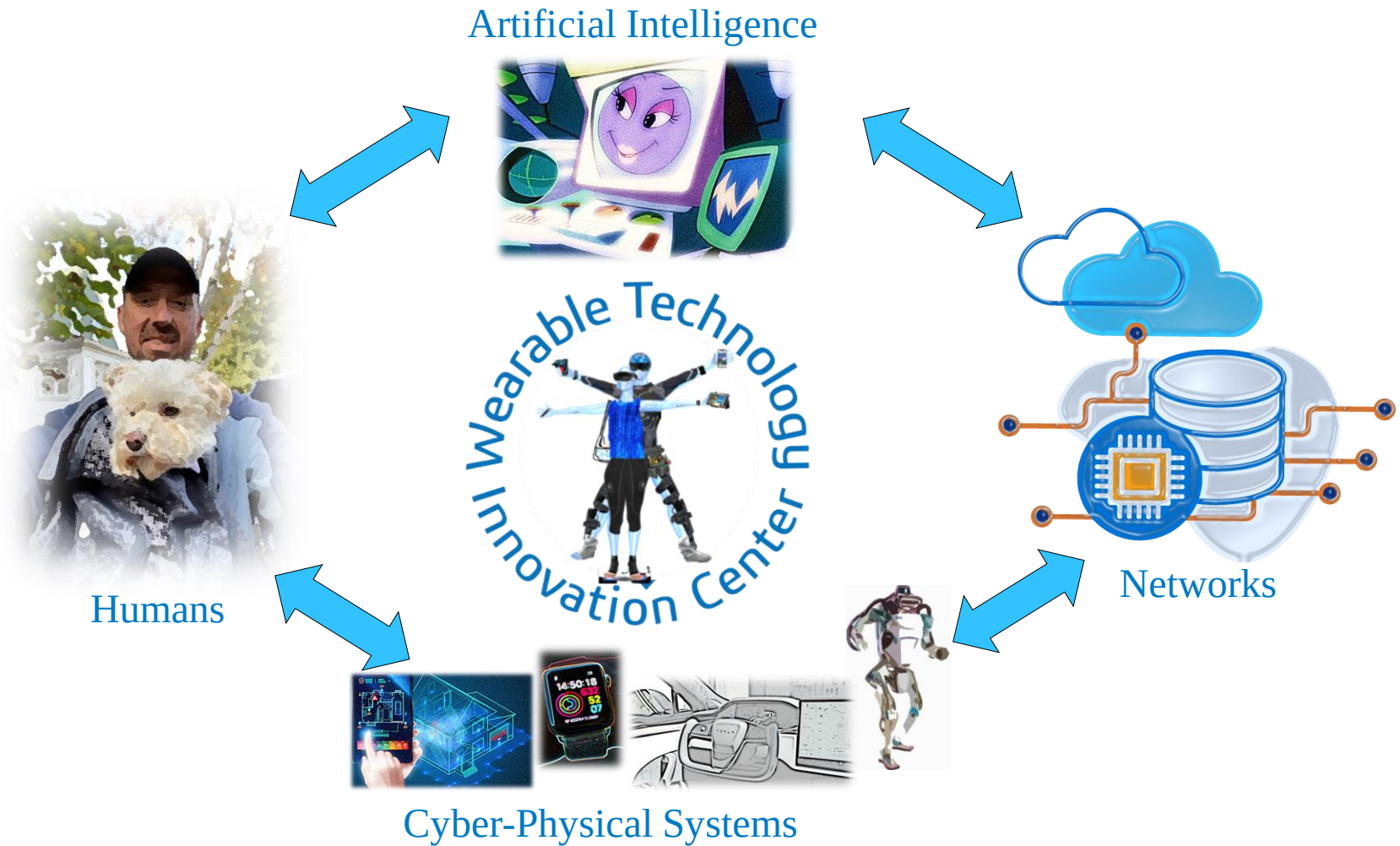


Stakeholders

- What are some other uses for this technique?
 - Think about other groups...
- Medical patients
- Caregivers
- Parents
- Doctors
- Trainers
- Therapists (PT/OT)
- Nurses
- Children
- Government agencies
- Early childhood teachers
- Athletes
- Coaches/Teams
- Safety engineers
- Researchers
- Businesses
- Tech companies
- Non-profit agencies
- Athletic agencies
- Etc...

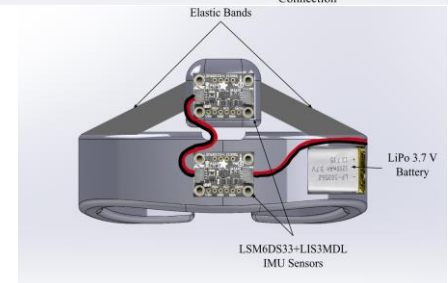
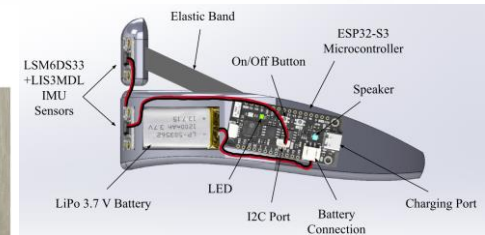
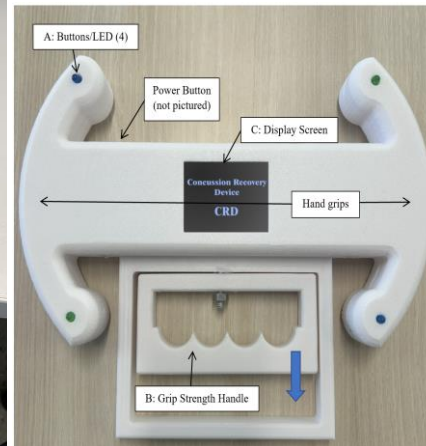
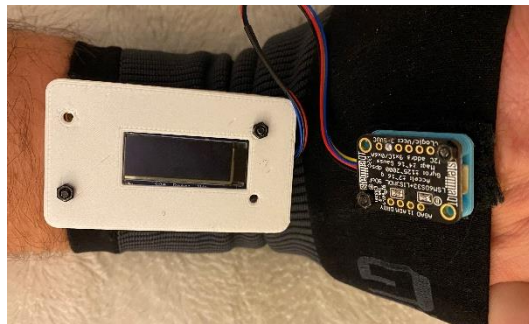
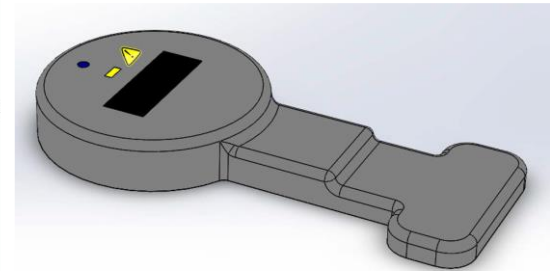
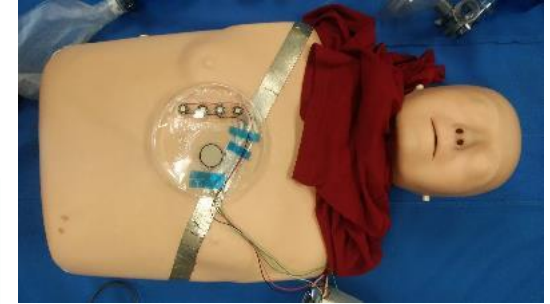
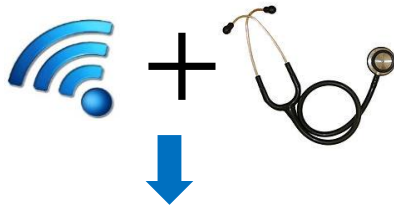
Boswell, M.A., Kidziński, L., Hicks, J.L. *et al.* Smartphone videos of the sit-to-stand test predict osteoarthritis and health outcomes in a nationwide study. *npj Digit. Med.* **6**, 32 (2023).

Collaborating for Enhancement



<https://www.ltu.edu/wtic>

Wearable Tech Capstone Projects



“if ever man fell ill, there was no defense—no healing food, no ointment, nor any drink—but for lack of medicine they wasted away, until I showed them how to mix soothing remedies with which they now ward off all their disorders.”

-Prometheus Bound, Aeschylus 525-455 BC

